



United International University

Department of Computer Science & Engineering
Mid-Term Exam, Trimester: Spring 2025

Course Code: CSE 3715
Duration: 1 hour 30 Minutes

Course Title: Data Communication
Total Marks: 30

(Any examinee found adopting unfair means would be expelled from the trimester/program as per UIU disciplinary rules.)

Answer all the questions

- (a) **Describe** the key components required for a successful communication between devices. [3]
(CO1)

(b) **Discuss** the advantages and disadvantages of mesh topology. [2]
(CO1)
- Show** the contents of the packets and frames at the network and data link layers for each hop as the data is sent from Device A to Device B in Figure 1. [6]
(CO1)

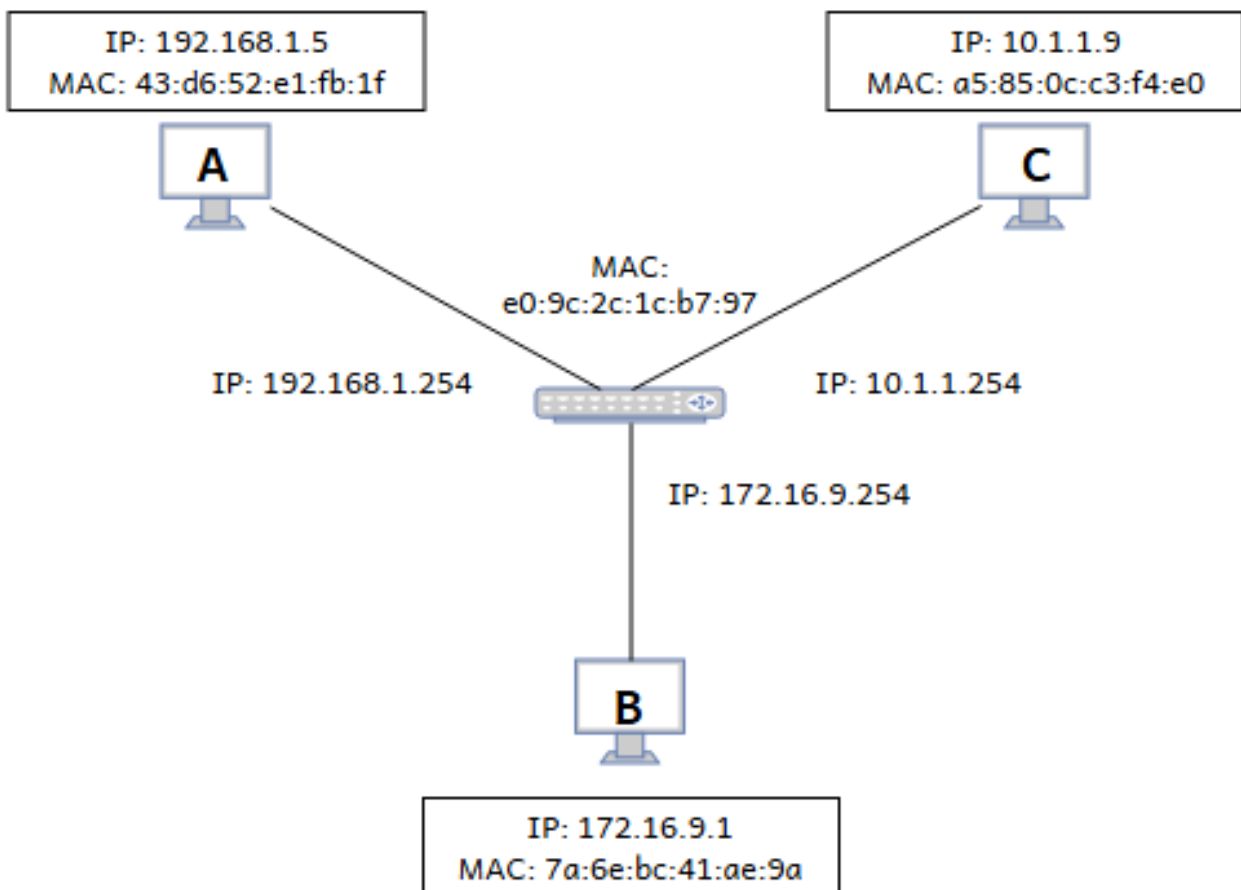


Figure 1: A Network Model

3. (a) A composite signal is composed of the following three waves:

- (i) $f_1(t) = 100 \sin(50\pi + 90)$
- (ii) $f_2(t) = 200 \sin(150\pi - 90)$
- (iii) $f_3(t) = 300 \sin(200\pi + 180)$

Determine the frequency of each wave and **calculate** the bandwidth of the composite signal.

[3]
(CO2)

(b) A communication channel has a bandwidth of 2 MHz and a signal-to-noise ratio (SNR) of 127. **Calculate** the maximum theoretical bit rate and **determine** the minimum number of signal levels required for transmission.

[4]
(CO2)

4. (a) **Explain** the DC component problem in line coding and **illustrate** it with a diagram.

[2]
(CO3)

(b) **Convert** the bit stream '**101100**' into digital signals using the following schemes:
(Assume the last signal was negative)

- (i) NRZ-I
- (ii) Polar
- (iii) Manchester

[6]
(CO3)

5. (a) **Explain** why Quadrature Amplitude Modulation (QAM) is used.

[2]
(CO4)

(b) **Illustrate** with a diagram how the data '**100101**' is converted using binary amplitude shift keying (ASK) and phase shift keying (PSK).

[2]
(CO4)